Chapter 22

NICE Guidance and Health Technology Assessment

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Learning Objectives:

Objective 22.1 Overview of National Institute for Health and Care Excellence (NICE) guidance development and evidence-based medicine.
Objective 22.2 Explain the purpose, production, and principles of technology appraisal guidance.
Objective 22.3 Detail the NICE technology appraisals process and decision-making.

OBJECTIVE 22.1. OVERVIEW OF NATIONAL INSTITUTE FOR HEALTH AND CARE EXCELLENCE GUIDANCE DEVELOPMENT AND EVIDENCE-BASED PRACTICE

National Institute for Health and Care Excellence (NICE) is a world-renowned organisation and its guidance and guidance production processes are held in high regard internationally.[1] NICE was originally set up in 1999 as the National Institute for Clinical Excellence to reduce variation in the availability and quality of National Health Service (NHS) treatments and care in England. This was primarily achieved through the appraisal of individual health technologies and the development of clinical guidelines. In 2005, after merging with the Health Development Agency, NICE began to develop public health guidance to help prevent ill health and promote healthy lifestyles. In 2013, NICE took on responsibility for developing guidance and quality standards in social care.

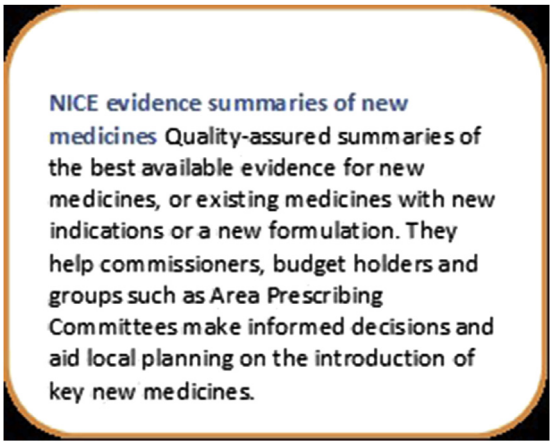
NICE is accountable to the Department of Health (the governmental department responsible for health), but operationally the Institute is independent of government. Independent committees are responsible for the development of key NICE guidance and recommendations. Before the establishment of NICE, though mainly at the margins, health authorities and the General Practice fundholders made differing purchasing decisions for healthcare. This resulted in variation in access to certain treatments for patients based on their healthcare provider catchment area—the so-called "postcode lottery." Over time NICE's recommendations have, to a great extent, resolved this problem. The establishment of NICE has also led to the faster uptake of innovative products, not just by the leading treatment centers but nationally.

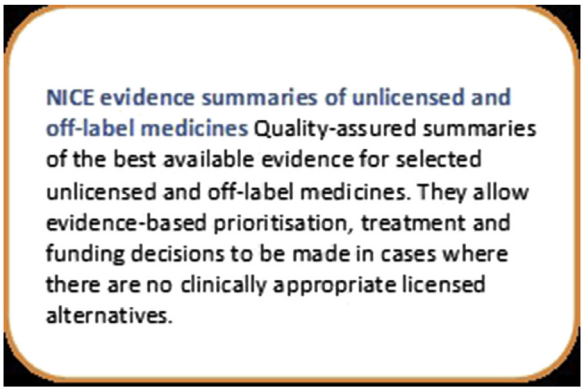
Evidence-based medicine (EBM) was adopted in the NHS for many years and guided clinical decision-making long before the establishment of NICE. A widely accepted definition states "EBM is the conscientious, explicit and judicious use of current best practice in making decisions about the care of individual patients."[2] The practice of EBM involves integrating individual clinical expertise with the most up-to-date and relevant available clinical evidence from systematic research. In addition, patients should be involved in decisions about their care.

The National Institute for Health and Care Excellence (NICE) provides national guidance and advice in England to improve health and social care.

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Another challenge today is that, with the growth of online resources, healthcare professionals and patients are overloaded with information from many different sources. It is impossible to stay abreast of publications and innovations even in a narrow field. From the information that health professionals actually consider, they need to ensure that it is appropriate and of high quality. Specific research findings may not always be the best source of evidence to inform a particular clinical decision, and there are many areas of medicine where research has not been carried out or is sparse. In addition, trial reporting transparency remains suboptimal despite great advances made in this field over the recent years.[3] NICE helps tackle this problem via NICE Evidence Services that enable access to authoritative clinical and nonclinical evidence and best practice to help professionals from across the NHS, public health and social care sectors make better decisions. The service also works directly with professionals and practitioners to identify evidence and to support the uptake and use of evidence to improve practice and care for people using services. Worldwide, there are many initiatives and organizations helping systematize and organize existing and emerging clinical knowledge (Cochrane,[4] registries, disease- and drug-specific databases, etc.).





NICE is an advisory body to the NHS. This means that NICE guidance is not mandatory, but recommendatory in nature. However, for decisions made by the technology appraisal (TA) committees on pharmaceuticals and selected medical devices, a positive recommendation is subject to mandatory funding by the NHS. The NHS Constitution states that patients have the right to drugs and treatments that have been recommended by NICE for use in the NHS, if their doctor believes they are clinically appropriate. Thus, medical professionals are advised to follow NICE guidance but make decisions regarding the treatment of each patient according to patient's condition (e.g., existing comorbidities, fitness level, ability to tolerate treatments), ongoing treatments, unmet need, and best clinical judgment in each given situation.

NICE Guidance

NICE produces six types of guidance which represent evidence-based recommendations:

- guidelines covering clinical topics, medicines practice, public health and social care
- diagnostics guidance
- highly specialized technology guidance
- interventional procedures guidance
- medical technologies guidance
- technology appraisals guidance

All guidance is developed by independent Committees. Draft guidance or interim documents (such as the scope) are consulted on. The consultation process includes registered stakeholders such as competitor companies, professional bodies (e.g., Association of British Pharmaceutical Industries (ABPI), Royal Colleges), patients and patient organizations, clinicians and members of the public (who can register to participate in the consultation process). NICE may also publish a range of supporting documents for each piece of guidance, including advice on how to put the guidance into practice, and on its costs and the evidence it is based on. Tools to support the local implementation of guidance include costing tools or statements and clinical audit tools.

In addition to guidance, NICE also produces quality standards which are a set of specific, concise statements and associated measures. They set out aspirational, but achievable, markers of high-quality, cost-effective patient care, covering the treatment and prevention of different diseases and conditions. Quality standards are based on guidance and advice from NICE and other organizations using NICE-accredited processes.[5] Quality standards help clinicians and organizations improve the quality of care they provide or commission.

Guidance production at NICE follows specific transparent processes and principles. Each guidance program has methods and process guide which can be found on the NICE website.[6] All information relating to the production of guidance (besides commercially sensitive data) is published on the NICE website. NICE committee meetings are being held in public and anyone can register to attend them. All NICE guidance is subject to a review process either at a set time (e.g., every 3 years) or when new evidence emerges (this can be identified by the Institute or notified by any stakeholder). NICE decisions can be appealed. Scientific rigor underpins all guidance production programs, and Committees are presented with the best available systematically identified evidence, which is reviewed by independent academic assessors. Finally, all decision-making at NICE is underpinned by the principles of equality and diversity. NICE's equality scheme describes how the Institute meets these commitments and obligations.[7] The Appraisal Committees also take into account the Institute's guidance on social value judgments described in the Institute's document, "Social value judgments: principles for the development of NICE guidance."[8]

OBJECTIVE 22.2. EXPLAIN THE PURPOSE, PRODUCTION, AND PRINCIPLES OF TECHNOLOGY APPRAISAL GUIDANCE

Technology Appraisals (TAs)

TAs at NICE assess the clinical and cost-effectiveness of health technologies. This is a rigorous process facilitating the decision on the adoption of innovative healthcare technologies into NHS clinical practice. TA recommendations can be issued on

- medicines
- medical devices
- diagnostic techniques
- surgical procedures
- health promotion activities.

The underlying methodology for TA is rooted in the principles of health technology assessment (HTA), which is an independent research about the effectiveness, costs, and broader impact of healthcare (treatments and tests). The reimbursement decision at NICE is informed by cost-utility analysis, with quality-adjusted life years (QALYs), used to measure the benefits of healthcare technologies, which is a key methodological feature of the NICE appraisals.

By combining improvements in survival and quality of life (QoL), QALYs are a uniform unit of benefits, providing a measure that can be used across all types of health interventions—whether a new diagnostic, medicine, surgical, or psychological treatment. The use of QALYs allows the comparison of the value of one treatment against another, leading to consistent decision-making across interventions and disease areas. New interventions are compared to established care, which is standard therapy currently available in the NHS in the given condition. Established care represents the comparator. To estimate QALYs gained with an intervention, it is essential to collect data on QoL and duration of life for patients in the intervention group and in the comparator group. QoL is a number between 0 and 1, called utility, with 1 representing perfect health and 0 representing death (though health states worse than death are possible). There are many generic QoL instruments (e.g., SF-36, SF-D6, EQ-5D, HUI2, HUI3) that measure utilities and many more disease-specific QoL instruments. NICE's preferred instrument is the EQ-5D, which is a standardized five-dimensional instrument. It is completed by the person (or a carer) having a treatment and is easy to use.

Cost-utility analysis - one of the tools used to carry out an economic evaluation. The benefits are assessed in terms of both quality and duration of life, and expressed as quality-adjusted life years (QALYs)

Quality-adjusted life year (QALYs) is a measure of the state of health of a person or group in which the benefits, in terms of length of life, are adjusted to reflect the quality of life. One QALY is equal to 1 year of life in perfect health. QALYs are calculated by estimating the years of life remaining for a patient following a particular treatment or intervention and weighting each year with a quality-of-life score (on a 0 to 1 scale). It is often measured in terms of the person's ability to carry out the activities of daily life, and freedom from pain and mental disturbance.

The cost-effectiveness of a new intervention is established by calculating the incremental QALYs associated with its use, compared with established practice, and the incremental costs. This allows the determination of the incremental cost-effectiveness ratio (ICER), which is the ratio of the difference in costs to the difference in QALYs. The ICER is a cost per QALY, and intuitively it can be understood that an intervention associated with a lower ICER (lower cost per QALY) is to be preferred to one with a higher ICER (greater cost per QALY). However, the appraisal of an individual healthcare technology is not a comparison with other available new technologies. Rather the cost-effectiveness (or ICER) of the technology is compared with a threshold.

The reason that decision-makers are faced with such difficult choices is because NHS in England operates within a fixed budget determined by parliament. The fixed budget necessitates that when a new treatment is recommended, the availability of some currently accessible therapy within the health system has to be withdrawn to fund it.

The concept of the opportunity cost underpins the idea of cost-utility analysis-based decision-making. The opportunity cost of an intervention means that investing in it displaces other treatments, diagnostics, or healthcare interventions somewhere within the system. As long as the cost-effectiveness of the treatment being displaced is worse (higher ICER) compared with what is being adopted, more health (as measured by QALYs) is being gained than what is being foregone, and the system is made more efficient, that is, more QALYs are generated for the same spend.

In actual fact, it is not possible to determine what treatments are, or should be displaced, by the adoption of a new technology. Therefore, NICE uses a threshold of £20,000 to £30,000 per QALY, to decide on whether a new technology is value for money. The threshold represents the cost-effectiveness of the interventions being displaced by the new technology and, in a rational system, should also represent the cost-effectiveness of the least cost-effective treatment currently available in the health system. The threshold, therefore, relates to the opportunity costs, in terms of health, of what is being displaced within the health system, given that it operates within a fixed budget. The threshold is not the willingness to pay for a QALY or the worth of a year of good health.

Methods

TAs, as with many other evaluations at NICE, are based on the PICO (population, intervention, comparator, and outcome) framework. It is a structured approach for developing review questions that divides each question into four components: the population (the population being studied); the interventions (what is being done); the comparators (other main treatment options in established clinical practice); and the outcomes (measures of how effective the interventions have been). The technology being appraised is listed in the scope of an appraisal under ‘intervention’. NICE can only make recommendations about the intervention being appraised. A comparator technology is one that is currently used in the NHS and could be replaced by the intervention, if recommended. NICE cannot issue guidance or make recommendations about comparator technologies (unless also listed as an intervention in a multiple technology appraisal [MTA]).

NICE has defined a “reference case” that specifies the elements of HTA and methods considered, by the Institute, to be appropriate for the Appraisal Committee’s purpose, and consistent with the objective of maximizing health gain from limited resources.[9] Submissions to the Institute should include an analysis generated using the reference case methods. This does not preclude additional analyses being presented, where one or more aspects of methodology differ from the reference case. However, these must be justified and clearly distinguished from the reference case analysis. Technology Appraisal Committees base their recommendations on a review of clinical and economic evidence. Clinical evidence shows how well the medicine or treatment works. Economic evidence shows how well the medicine or treatment works in relation to how much it costs the NHS—does it represent value for money? The perspective for evaluation of healthcare technologies at NICE is that of the NHS and PSS (Personal Social Services) for costs. This means that the submission should include only resource use and costs associated with the use of the intervention and comparator which are borne by the NHS and PSS and not, for example, by the society or patients themselves. The perspective on outcomes is all direct health effects, whether for patients or, when relevant, carers.

Economic evaluation requires setting a time horizon, which is the period over which the main differences between interventions in benefits and the use of resources in health and social care are expected to be experienced. In most instances, a lifetime horizon (modeling all patients to death) is adopted, which means that the assumptions on the long-term clinical effectiveness of intervention and other parameters are required. This also increases uncertainty for the decision-makers.

Types of Appraisals and Processes

The decision on which process will be used to appraise a technology is made during the topic selection. TA process at NICE has different formats:

- STA - a single technology appraisal which covers one technology for one indication.
- MTA - a multiple technology appraisal, which covers more than one technology for a single indication or one technology for more than one indication (under 10% of all appraisals).[10]
- FTA - a fast-track appraisal, which is a faster version of an STA for highly cost-effective new treatment. This process was launched on April 1, 2017.[11]

The STA process is specifically designed to appraise a single product, device, or other technology for a single indication. The process normally covers new technologies (typically, new pharmaceutical products or newly licensed indications) and enables NICE to produce guidance soon after the technology is introduced in the United Kingdom. NICE seeks relevant evidence from several sources. The company submits the principal evidence. Consultees provide information and selected clinical experts, NHS commissioning experts, and patient experts also give evidence. The Evidence Review Group (ERG), an external academic organization independent of NICE, produces a review of the submitted evidence.

The MTA process is designed to appraise single or multiple products, devices, or other technologies with one or more related indications. NICE seeks relevant evidence from several sources. An independent academic group, the Assessment Group, carries out a systematic review of the evidence and economic analysis. Consultees provide information and selected clinical experts, NHS commissioning experts, and patient experts also give evidence. An overview of the TA development process is presented in Fig. 22.1. The range of participants in a NICE TA process is presented in Fig. 22.2.

The Highly Specialised Technologies (HST) program only considers drugs for very rare conditions (most orphan products, as classified by regulators, are assessed via the TA program at NICE). The methods for the HST program vary from TAs in some instances and can be reviewed on the NICE website.[12]

The majority of topics for the HST program are identified by the National Institute for Health Research Innovation Observatory. They aim to notify the Department of Health of key, new, and emerging healthcare technologies that might need to be referred to NICE against the following time frames:

- new drugs, in development, at 20 months to marketing authorization
- new indications, at 15 months to marketing authorization.

A single HST evaluation can only cover a single technology for a single indication. Cost-utility analysis is required for an HST committee to make a decision.

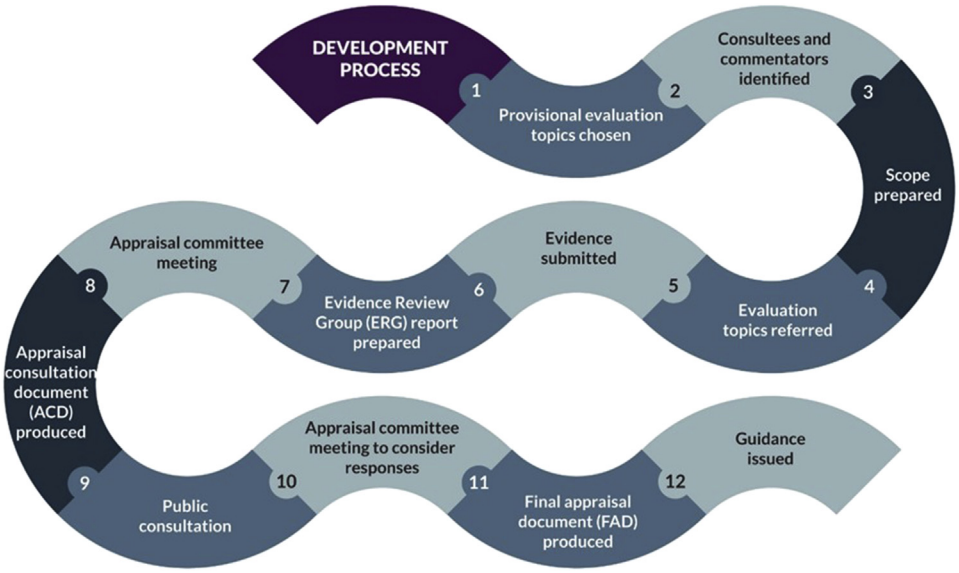


FIGURE 22.1 Overview of NICE technology appraisal development process.

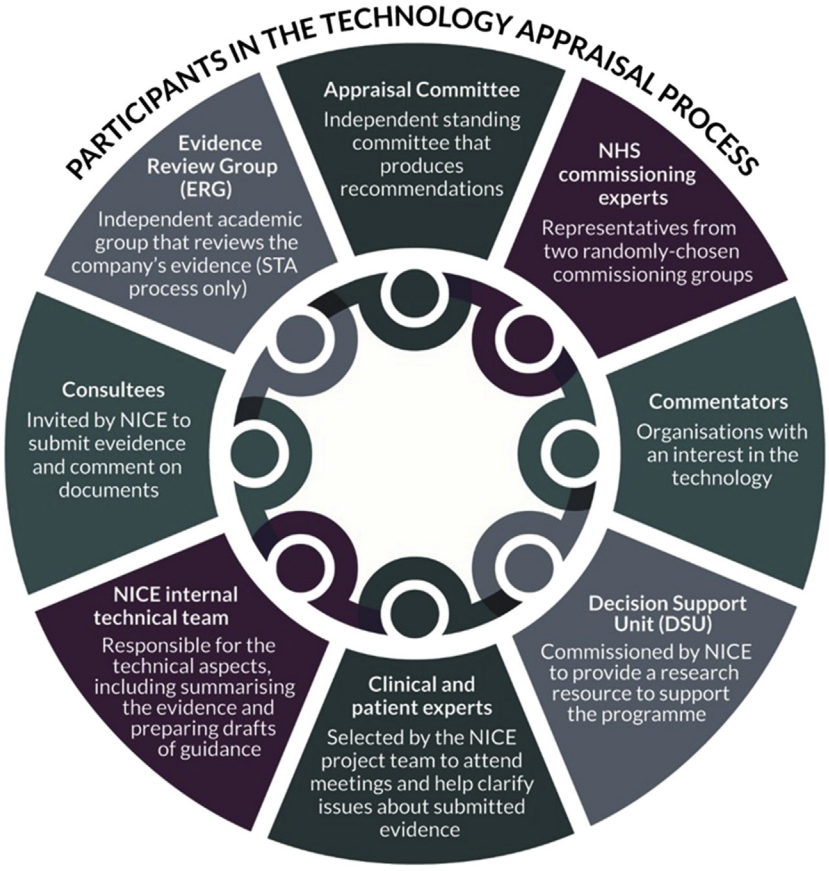


FIGURE 22.2 Range of participants in a NICE technology appraisal process.

OBJECTIVE 22.3. DETAIL THE NICE TECHNOLOGY APPRAISAL COMMITTEE PROCESS AND DECISION-MAKING

Members of a NICE TA Committee are appointed for a 3-year term and are drawn from

- the NHS
- patient and carer organizations
- academia
- pharmaceutical and medical devices industries

The Appraisal Committee considers the evidence and makes a judgment on whether or not the technology should be recommended as a clinically effective and cost-effective use of NHS resources or whether it should only be recommended for specific subgroups of people. The Appraisal Committee submits its recommendations to NICE in either an appraisal consultation document (ACD) or a final appraisal determination (FAD). The Appraisal Committee produces an ACD only if its preliminary recommendations are substantially more restrictive than the terms of the marketing authorization (or equivalent, for example, CE marking for devices) of the technology being appraised or do not recommend the use of the technology. If the Committee produces an ACD, then NICE invites consultees, commentators, and the public to comment on the ACD. After considering these comments, the Committee finalizes its recommendations and submits them to NICE in the form of a FAD. The FAD forms the basis of the guidance that NICE issues to the NHS in England.

The estimation of cost-effectiveness (based on an ICER) is not the only criterion for decision-making, and qualitative criteria such as unmet medical need, patient and clinician's testimony, equity principles, social value judgments are used in deliberations to arrive at the final decision. Below a most plausible ICER of £20,000 per QALY gained, the decision to recommend the use of a technology is normally based on the cost-effectiveness estimate. Above a most plausible ICER of

£20,000 per QALY gained, judgments about the acceptability of the technology as an effective use of NHS resources will specifically take account of the following factors:

- The degree of certainty around the ICER
- Whether there are strong reasons to indicate that the assessment of the change in health-related QoL has been inadequately captured
- The innovative nature of the technology
- The technology meets the criteria for special consideration as a "life-extending treatment at the end-of-life"
- Aspects that relate to nonhealth objectives of the NHS

Above a most plausible ICER of £30,000 per QALY gained, the Committee will need to identify an increasingly stronger case for supporting the technology as an effective use of NHS resources, with regard to the factors listed above. In some situations, at the end-of-life, the Committee can accept a higher ICER of up to £50,000/QALY. For interventions evaluated via the HST program, a higher threshold of £100,000−£300,000/QALY is considered, depending on the QALY gain associated with the intervention.

TA decisions may contain more than one recommendation. NICE classifies its recommendations into five categories:

- Recommended
- Optimized (recommended for only a subgroup of the licensed population)
- Only in research
- Not recommended
- Recommended for use in the CDF (Cancer Drug Fund).[13]

For all NICE appraisals, the Institute acts as a "price taker," with the cost of the health technology determined by the manufacturer. However, where the manufacturer's price results in the intervention not being cost-effective, the manufacturer has the option of offering a pricing agreement designed to improve cost-effectiveness and facilitate patient's access, through a Patient Access Scheme.[14] These discounts are negotiated between the manufacturer and the Department of Health. NICE only ensures that they are operationally viable.

CONCLUSION

NICE issues various types of evidence-based guidance, each of which may be considered specific instances of HTA. Transparent processes have been developed to include various stakeholders, who have defined roles and responsibilities. Cost-effectiveness is estimated using consistent methodology according to a reference case. The decisions are made by the independent Committees. The cost per QALY is compared against a threshold, and the Committee takes into account additional qualitative factors. Technologies for very rare disease are evaluated by a separate program, using a higher threshold. A patient access scheme may be proposed by the manufacturer to lower the effective cost of the technology, improve cost-effectiveness, and improve patients' access to innovative therapies.